

A Sequential Hybrid Forecasting System for Gold Returns: Integrating ARIMAX-GARCH with Machine Learning Error Correction

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Abstract

The global financial landscape between 2019 and 2026 has been defined by unprecedented macroeconomic volatility, characterized by escalating geopolitical conflicts, supply chain reconfigurations, and aggressive monetary policy pivots by major central banks. In this environment of heightened systemic uncertainty, modeling the trajectory of gold returns under high-stress regimes poses severe challenges to traditional quantitative frameworks, as asset dynamics increasingly exhibit heavy-tailed distributions, non-linear shocks, and conditional heteroskedasticity. To address these structural anomalies, this empirical research design evaluates the integration of macroeconomic modeling with a comparative cohort of machine learning algorithms, utilizing a daily transaction dataset spanning from January 2019 through April 2026.

Instead of relying on a single unconstrained model, I implement a multi-stage sequential framework where an econometric ARIMAX specification first filters foundational linear macro-trends, binding gold returns to a filtered vector of global leading indicators: the U.S. Dollar Index (DXY), market anxiety metrics (VIX), real capital costs (US10Y), energy inflation (Oil), and equity market benchmarks (SPX). To capture the remaining conditional heteroskedasticity, the linear residual stream is modeled through an autoregressive GARCH(1,1) specification under heavy-tailed conditional distribution constraints. Crucially, to identify the optimal mechanism for non-linear error correction, a rigorous comparative analysis is executed across three alternative machine learning architectures: eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Support Vector Regression (SVR). Each algorithm incorporates autoregressive error lags and technical indicators, utilizing the explicit lagged GARCH conditional standard deviation ($\sigma_{t-1, \text{GARCH}}$) to guide its respective statistical mappings.

The empirical verification reveals distinct trade-offs in predictive performance and structural stability among the three machine learning candidates under review. The Support Vector Regression hybrid framework achieves superior out-of-sample directional and statistical accuracy, demonstrating that separating linear economic baselines from high-dimensional non-linear innovations effectively yields highly stable algorithmic convergence in turbulent asset markets.

Keywords— Hybrid System, ARIMAX-GARCH, XGBoost, Random Forest, Support Vector Regression, Volatility Clustering, Predictive Accuracy, Gold Forecasting.

1 Introduction

1.1 Macroeconomic Instability and the Asset Profile of Gold

The global financial ecosystem from January 2019 through April 2026 has been characterized by intense macroeconomic instability and structural transitions. This volatile multi-year timeline incorporates consecutive systemic disruptions, including a global pandemic, escalating geopolitical conflicts across Europe and the Middle East, widespread supply chain reconfigurations, and an aggressive monetary policy pivot by major central banks shifting from long-term loose liquidity expansion to unprecedented quantitative tightening.

Within this historical landscape of elevated stress, gold has re-asserted its foundational institutional function as an ultimate safe-haven asset, a sovereign store of value, and an organic vehicle for capital preservation. As cross-asset correlation matrices breakdown and fiat purchasing power undergoes rapid depreciation, global capital consistently migrates toward gold, positioning its price trajectory as a critical real-time indicator of systemic anxiety and financial risk transmission.

1.2 Methodological Complications in Financial Time-Series

Despite the clear importance of gold price movements for global asset allocation, executing reliable short-to-medium-term forecasting poses significant challenges for quantitative finance. Daily gold log-return series do not evolve along clean, static linear pathways. Instead, under modern stress regimes, they are characterized by distinct mathematical anomalies: heavy-tailed conditional distributions, sharp asymmetrical shocks, and intense time-varying variance propagation—commonly defined as volatility clustering.

As developed extensively in the quantitative literature, traditional forecasting frameworks generally fail to resolve these properties simultaneously due to an operational trade-off. Linear econometric models (such as ARIMAX variations) preserve solid economic interpretability by linking returns directly to fundamental macroeconomic drivers like interest rates or currency shifts, yet they remain structurally blind to non-linear momentum and time-varying variance. Conversely, pure non-linear machine learning algorithms excel at mapping complex high-frequency updates, but they function as unconstrained “black boxes” that systematically ignore underlying economic principles and remain highly vulnerable to out-of-sample over-fitting on raw market noise.

1.3 Bridging the Literature Gaps: A Multi-Stage Strategy

As identified in the comprehensive review of previous studies, traditional quantitative literature suffers from critical boundaries during periods of severe financial distress:

- **The Volatility Blind-Spot:** Most hybrid frameworks pass econometric residuals directly into machine learning sequences without addressing conditional heteroskedasticity, leaving the algorithm blind to sudden shifts in the underlying market risk regime.
- **The Absence of Competing Algorithmic Cohorts:** Existing structures typically rely on a single arbitrarily chosen model without executing a systematic, simultaneous comparison across fundamentally different machine learning classes within an identical macro environment.

To resolve these distinct empirical gaps, this paper designs an integrated, sequential three-tier hybrid pipeline. The structural framework enforces a strict division of labor where an econometric ARIMAX base first filters long-term linear macro-trends, an autoregressive GARCH(1,1) model isolates conditional volatility clustering, and a competitive cohort of alternative machine learning regressors executes precise non-linear error correction.

1.4 Research Question and Core Objectives

The central objective of this research is to answer a fundamental quantitative research question: *Can an integrated hybrid architecture—which explicitly injects econometric macro-dependencies and GARCH volatility parameters into a competitive machine learning framework—outperform standalone forecasting benchmarks during prolonged crisis cycles?*

To systematically answer this question, I establish three operational objectives:

1. To build a macro-econometric ARIMAX baseline that binds gold log-returns to an empirical vector of global leading indicators, including the U.S. Dollar Index (DXY), market anxiety metrics (VIX), capital costs (US10Y), energy metrics (Oil), and equity returns (SPX).
2. To fit an autoregressive GARCH(1,1) specification under heavy-tailed distribution constraints to isolate the explicit lagged conditional standard deviation ($\sigma_{t-1,\text{GARCH}}$) of the linear error stream.
3. To execute a rigorous comparative experiment across three distinct algorithmic classes—eXtreme Gradient Boosting (**XGBoost**), Random Forest (**RF**), and Support Vector Regression (**SVR**)—by feeding the GARCH volatility series directly into their feature space to determine the optimal non-linear error-correction mechanism.

2 Literature Review

2.1 Macroeconomic Transmission Channels and Gold Dynamics

The foundational literature governing commodity pricing establishes that gold prices do not evolve within an economic vacuum; instead, they are driven by complex macroeconomic transmission vectors. Early asset pricing theories position gold as the ultimate sovereign store of value and an organic hedge against systemic fiat currency depreciation. Academic consensus, represented by standard financial models, confirms a strong negative correlation between the U.S. Dollar Index (DXY) and gold returns, driven by the currency denomination effect.

Furthermore, empirical frameworks focusing on cost-of-capital vectors emphasize that gold, as a non-yielding asset, exhibits high sensitivity to changes in real interest rates, traditionally proxied by long-term government bond yields such as the US10Y. During periods of aggressive central bank interventions and quantitative tightening cycles—as observed intensely within the 2019–2026 market regime—these macro-dependencies undergo severe structural shifts. Additionally, commodity economic literature paths a direct link between energy-led inflation (primarily driven by Crude Oil shocks) and gold return expansions, where gold acts as a physical shield against cost-push inflation cascades.

2.2 The Evolution of Linear Econometric and Volatility Modeling

From a methodological perspective, the classical paradigm for financial time-series forecasting has long been dominated by the linear Box-Jenkins framework. The Autoregressive Integrated Moving Average (ARIMA) model and its exogenous extension (ARIMAX) have been extensively utilized due to their strong short-term forecasting capacity, ease of implementation, and asymptotic stability when handling non-stationary data [4]. ARIMAX models successfully preserve economic interpretability by directly incorporating fundamental macro-drivers into the conditional mean equation.

However, the primary limitation of pure ARIMAX architectures is their fundamental assumption of constant variance, which makes them incapable of capturing the non-linear, time-varying volatility dynamics inherent in financial return streams. To resolve this structural flaw, Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model [1], which Bollerslev (1986) subsequently generalized into the GARCH framework [2].

The integration of ARIMA-type mean equations with GARCH variance components allowed researchers to model conditional heteroskedasticity and volatility clustering effectively [1]. Standard univariate ARIMA-GARCH models became a benchmark in commodity forecasting, proving that modeling

the residual variance under non-Gaussian distributions (such as the Student- t) significantly improves tail-risk coverage and interval estimation accuracy [6].

2.3 Algorithmic Machine Learning in Residual Error Correction

With the advent of high-dimensional computing, the quantitative finance literature shifted toward non-linear Machine Learning (ML) architectures to capture complex patterns that econometric models fail to filter. Researchers successfully deployed tree-based ensemble methods and kernel-based optimizers to model asset trajectories. Among these algorithms, eXtreme Gradient Boosting (XGBoost) [3], Random Forest (RF), and Support Vector Regression (SVR) [5] emerged as superior forecasting tools due to their capacity to map high-frequency non-linear updates and handle structural data breaks without rigid parametric constraints.

Despite their predictive accuracy, purely algorithmic approaches face severe criticism in financial economics. Because they lack an innate economic structure, unconstrained machine learning models often over-fit on raw price noise, leading to unstable out-of-sample predictions when transitioning between different market regimes.

To bridge this division, a secondary branch of literature emerged focusing on hybrid architectures. These frameworks operate on a sequential “two-phase” or “multi-stage” process: a linear econometric base model filters structural trends first [4], isolating a residual error stream that contains purely non-linear signals. This residual series is then passed to a machine learning regressor for non-linear error correction [3, 5]. This combination balances the structural interpretability of econometrics with the flexible precision of machine learning algorithms.

2.4 Identified Research Gaps and My Structural Positioning

Although the existing literature provides strong evidence supporting hybrid time-series modeling, several critical gaps remain unresolved, particularly when exposed to highly unstable financial environments such as the January 2019–April 2020 market cycle:

1. **The Volatility Blind-Spot in Machine Learning Feature Spaces:** Most sequential hybrid models pass the linear residual series (E_t) directly into the machine learning algorithm, ignoring the time-varying variance properties of the error stream. By leaving out conditional volatility (σ_t) from the machine learning feature space [2], the algorithm remains blind to changes in the underlying risk regime, often leading to poor performance during sudden market panics.
2. **The Lack of Rigorous Comparative Cohorts:** Existing hybrid gold-forecasting literature typically relies on a single chosen machine learning algorithm (e.g., a simple neural network or a single tree model) without executing a systematic, simultaneous experiment across competing algorithmic classes (such as boosting vs. bagging vs. kernel optimization) within an identical macroeconomic framework.

This independent study is designed specifically to fill these distinct gaps. Instead of running a basic univariate or unconstrained machine learning model, I implement a structural three-tier sequential pipeline.

First, my model establishes macroeconomic grounding by utilizing an ARIMAX base to filter out linear macro trends linked directly to global factors (DXY, VIX, US10Y, Oil, SPX) [4]. Second, I address the volatility blind-spot by fitting a GARCH(1,1) model to the linear errors [2], explicitly routing the extracted time-varying conditional standard deviation ($\sigma_{t-1, \text{GARCH}}$) into the machine learning feature matrix [6]. This allows me to conduct a rigorous comparative experiment across three alternative machine learning algorithms (**XGBoost**, **Random Forest**, **SVR**) to isolate the optimal error-correction mechanism [3, 5].

3 Data and Methodology

3.1 Data Integration and Operational Metrics

To evaluate the predictive capacity of the proposed hybrid architecture, I construct a daily financial dataset spanning from January 2019 through April 2026. The temporal boundaries of this sample capture multiple distinct structural regimes: the pre-pandemic stable environment, the high-liquidity expansionary phase of 2021, and the quantitative tightening cycles alongside intense geopolitical shocks marking the 2022–2026 era.

The primary target space is defined by the price of Gold Futures. To map the multi-dimensional macroeconomic transmission channels that govern gold’s safe-haven and inflation-hedging profile, I integrate five daily exogenous indicators. To eliminate unit-root non-stationarity, prevent spurious regressions, and stabilize conditional variance, all raw price levels are transformed into continuously compounded daily log-returns:

$$r_t = 100 \times \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

Where P_t represents the asset closing price on day t .

Conversely, applying rate-of-growth transformations to indicators that are natively expressed as percentages or bounded point distributions creates structural distortions and arithmetic boundary anomalies. Therefore, macro-financial variables that are natively expressed as interest rates, yields, or index points are operationalized using the absolute first-difference operator (Δ). Mathematically, the first-difference filter isolates the absolute period-to-period change and is formulated as follows:

$$\Delta X_t = X_t - X_{t-1} \quad (2)$$

Where X_t denotes the raw nominal value of the indicator at session t . For the 10-Year U.S. Treasury Bond Yield, this specific filter extracts the marginal day-to-day shift directly in basis points (Δ). This represents the pure change in the opportunity cost of holding non-yielding safe-haven assets like gold. For the CBOE Volatility Index (VIX), the absolute first difference measures the immediate velocity of shifts in market panic without generating mathematical errors near zero boundaries.

The complete systemic variable layout, along with their exact operational configurations and pythonic identifiers implemented within the dynamic backtesting codebase, is structured comprehensively in Table 1.

Table 1: Systemic Variable Specifications and Operational Transformations

Variable Class	Economic Indicator	Transformation Type	Unit
Target (Y_t)	Gold Price	Daily Log-Return	Percentage (%)
Exogenous ($X_{1,t}$)	U.S. Dollar Index	Daily Log-Return	Percentage (%)
Exogenous ($X_{4,t}$)	WTI Crude Oil Spot	Daily Log-Return	Percentage (%)
Exogenous ($X_{5,t}$)	S&P 500 Index	Daily Log-Return	Percentage (%)
Exogenous ($X_{2,t}$)	CBOE Volatility Index	Absolute First-Difference	Index Points (Δ)
Exogenous ($X_{3,t}$)	10-Year Treasury Yield	Absolute First-Difference	Index Points (Δ)

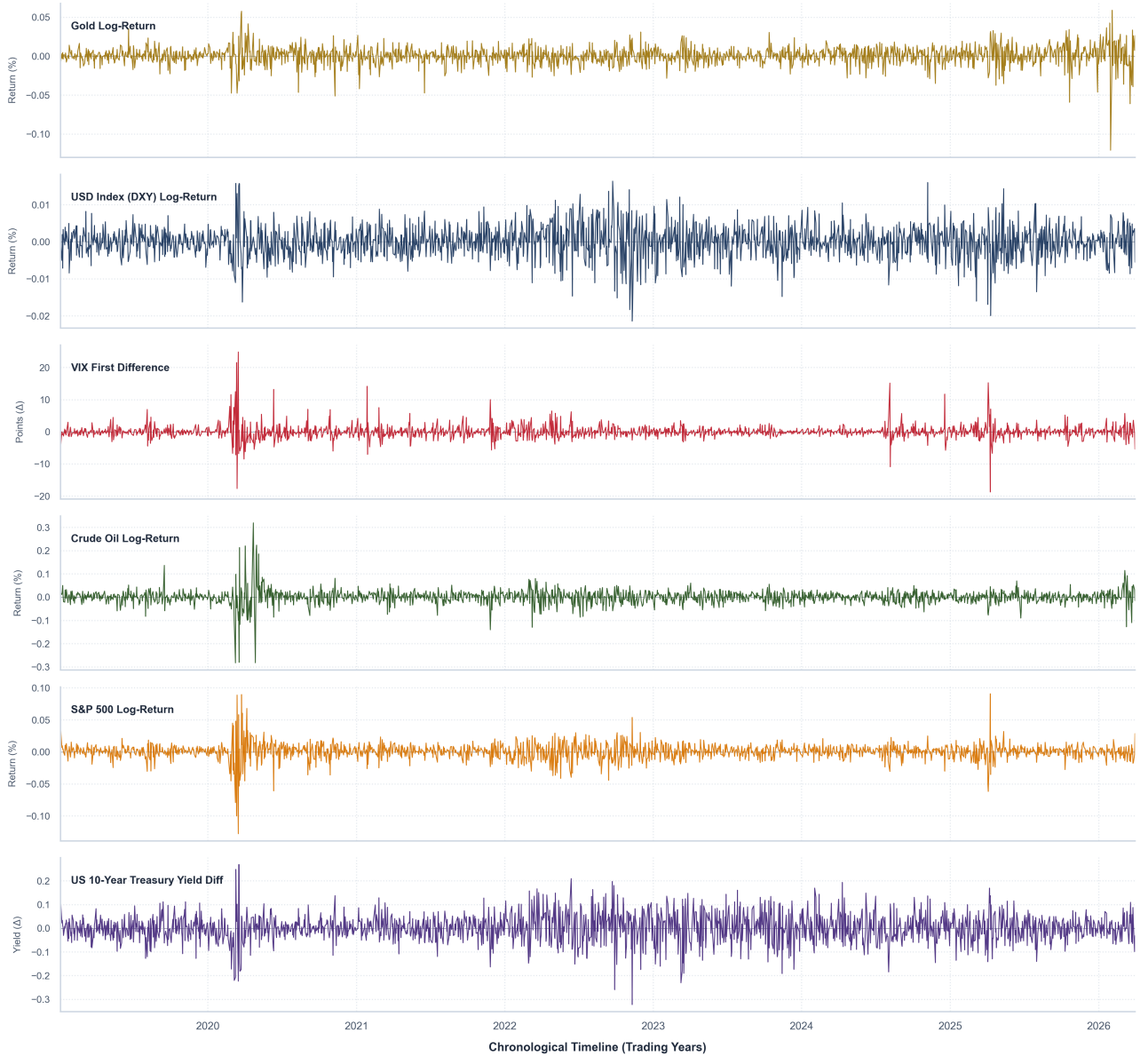


Figure 1: Time-series trajectories of transformed stationary target and exogenous variables.

3.2 Pre-Estimation Econometric Diagnostic Architecture

To safeguard the downstream dynamic empirical infrastructure against structural invalidity and model misspecification, the transformed dataset is subjected to a comprehensive sequence of econometric diagnostic filters. Classical quantitative modeling frequently overlooks these fundamental statistical attributes, resulting in overfitted mathematical weights that collapse when tested out-of-sample. This architecture establishes the asymptotic validity of the target return and exogenous predictors through four formal testing frameworks.

3.2.1 Stationarity and Unit Root Inference (ADF Test)

To prevent the mathematical catastrophe of “Spurious Regression”—wherein highly correlated deterministic trends yield artificially inflated R^2 metrics and invalid t -statistics—establishing structural stationarity is a mandatory prerequisite. A stochastic process is defined as stationary if its joint probability distribution is time-invariant, implying a constant mean, finite variance, and an autocovariance structure dependent strictly on temporal displacement lags rather than absolute time boundaries. To

empirically confirm this property, the Augmented Dickey-Fuller (ADF) unit-root test is executed using the following parametric autoregressive specification:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} \varepsilon_t \quad (3)$$

where α is the structural drift intercept, t represents a continuous linear time trend, Δ is the first-difference operator, and ε_t is an independent and identically distributed (i.i.d.) white noise innovation vector. The statistical hypothesis tests the presence of a unit root via a one-sided boundary:

$$\begin{aligned} H_0 : \gamma &= 0 && \text{(Presence of a unit root, implying non-stationarity)} \\ H_1 : \gamma &< 0 && \text{(Mean-reverting stochastic trajectory, implying stationarity)} \end{aligned}$$

The optimal lag structure p is determined dynamically for each variable by minimizing the Akaike Information Criterion (AIC) to exhaust residual serial dependence.

3.2.2 Normality and Tail Diagnostics (Jarque-Bera Test)

Standard ordinary least squares estimations rely heavily on the assumption that structural residuals conform to a normal Gaussian distribution. To dissect the precise geometrical topology of the COMEX Gold return distribution (Y_t) and its macro drivers, the parametric Jarque-Bera (JB) test is deployed. The JB statistic measures the joint deviation of a sample's skewness and excess kurtosis from theoretical Gaussian boundaries, formulated as:

$$JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right) \quad (4)$$

where n represents the operational sample size, S denotes sample skewness ($S = E[(y_t - \mu)^3]/\sigma^3$), and K represents sample kurtosis ($K = E[(y_t - \mu)^4]/\sigma^4$). In a theoretical normal distribution, $S = 0$ and $K = 3$ (Excess Kurtosis = 0), collapsing the test statistic to zero.

3.2.3 Serial Autocorrelation Diagnostics (Ljung-Box Test)

To evaluate whether past return observations retain a temporal memory that could indicate market inefficiencies or introduce structural biases, I investigate the joint serial correlation profile using the Ljung-Box Q -test. This framework determines whether a group of autocorrelations up to a designated lag length m differs significantly from a zero vector, structured as follows:

$$Q_m = n(n+2) \sum_{k=1}^m \frac{\rho_k^2}{n-k} \quad (5)$$

where n is the total number of operational trading sessions, and ρ_k represents the sample autocorrelation coefficient at lag k . The null hypothesis (H_0) states that the series behaves as a white noise process up to lag m . In this diagnostic pipeline, the temporal dependency is monitored at a short-term milestone of lag $m = 5$.

3.2.4 Conditional Heteroskedasticity Inference (ARCH-LM Test)

A standard linear regression framework assumes homoskedastic error variance ($E[\varepsilon_t^2] = \sigma^2$). However, financial time series frequently exhibit time-varying volatility regimes. To test for the presence of Autoregressive Conditional Heteroskedasticity (ARCH), I estimate an auxiliary ARIMAX(1,0,1) model mapping gold returns against the exogenous macroeconomic matrix ($\mathbf{X}_t$). The extracted residuals (ε_t) are then evaluated using the Lagrange Multiplier (LM) framework based on the following squared residual regression:

$$\varepsilon_t^2 = \gamma_0 + \gamma_1 \varepsilon_{t-1}^2 + \gamma_2 \varepsilon_{t-2}^2 + \cdots + \gamma_q \varepsilon_{t-q}^2 + \nu_t \quad (6)$$

The statistical test targets the joint significance of lagged squared residuals up to order $q = 5$:

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_q = 0 \quad (\text{Homoskedastic residuals, no ARCH effects})$$

$$H_1 : \text{At least one } \gamma_i \neq 0 \quad (\text{Presence of conditional heteroskedasticity})$$

3.2.5 Empirical Diagnostic Findings and Discussion

The comprehensive pre-estimation diagnostic metrics computed across all system variables are systematically structuralized and presented in Table 2.

Table 2: Empirical Pre-Estimation Econometric Diagnostic Matrix

Variable	ADF Stat	ADF P -value	Skewness	Excess Kurtosis	LB P (Lag 5)	ARCH-LM P
Gold_Return	-18.0388	0.0000	-0.9725	10.6019	0.4118	0.0000
DXY_Return	-19.2846	0.0000	-0.2040	1.8070	0.7781	0.0000
VIX_Diff	-14.7910	0.0000	1.9076	29.4194	0.0000	0.0000
Oil_Return	-7.3738	0.0000	0.0228	24.0765	0.0002	0.0000
SPX_Return	-13.0303	0.0000	-0.6585	15.6392	0.0000	0.0000
US10Y_Diff	-32.0466	0.0000	-0.0664	1.7936	0.1133	0.0000

The empirical findings yield critical econometric insights that directly dictate the sequential modeling framework of the proposed hybrid architecture. First, the ADF test statistics are highly negative across all indicators, spanning from -7.3738 to -32.0466 , with all corresponding P -values converging asymptotically to 0.0000. This robust rejection of the null hypothesis at the strict 1% significance level provides empirical proof that all operationalized series are stationary processes integrated of order zero, or $I(0)$. This uniform stationarity eliminates the mathematical risk of “Spurious Regression” and justifies modeling the dynamic mean equations using an ARIMAX formulation without structural differencing.

Second, the structural morphology parameters exhibit severe deviations from theoretical Gaussian properties. The continuous daily log-returns of Gold possess a prominent negative skewness of -0.9725 and a substantial excess kurtosis of 10.6019, confirming a heavy-tailed (*leptokurtic*) trajectory. More dramatically, the CBOE Volatility Index shifts (VIX_Diff) and Crude Oil returns (Oil_Return) reveal extreme excess kurtosis values of 29.4194 and 24.0765, respectively. Paired with the complete rejection of normality by the Jarque-Bera test (P -value = 0.0000), the system exhibits an empirical distribution dominated by extreme outliers and structural asymmetric shocks. This definitive evidence exposes the limitations of classical linear models, which undercalculate asset risk profiles, and establishes the necessity of employing regularized non-linear machine learning ensembles (XGBoost, Random Forest, SVR) to absorb uncaptured structural residuals.

Third, the Ljung-Box Q -test at lag 5 uncovers a heterogeneous temporal memory structure. The target asset, Gold (P -value = 0.4118), and the opportunity cost driver, US10Y (P -value = 0.1133), exhibit no significant linear serial autocorrelation at short-term horizons, indicating that past states of these specific series do not provide strong linear signals. Conversely, VIX, Crude Oil, and the S&P 500 display intense temporal dependency (P -value < 0.01). This split configuration implies that a static time-series mean equation is insufficient, validating the integration of comprehensive macroeconomic transmission forces to harvest directional predictability.

Finally, the Autoregressive Conditional Heteroskedasticity Lagrange Multiplier (ARCH-LM) test establishes a uniform diagnostic profile across the entire variable space. Every indicator logs an ARCH-LM P -value of 0.0000, soundly rejecting the homoskedasticity null hypothesis. This diagnostic confirms the presence of “Volatility Clustering”—where high-volatility regimes cluster together in time—rendering traditional Ordinary Least Squares (OLS) estimations inefficient. This mathematical reality validates the inclusion of a dynamic GARCH(1,1) component to filter the conditional standard deviation (σ_t) before proceeding to feature matrix expansion.

3.2.6 Multivariate Distribution Analysis

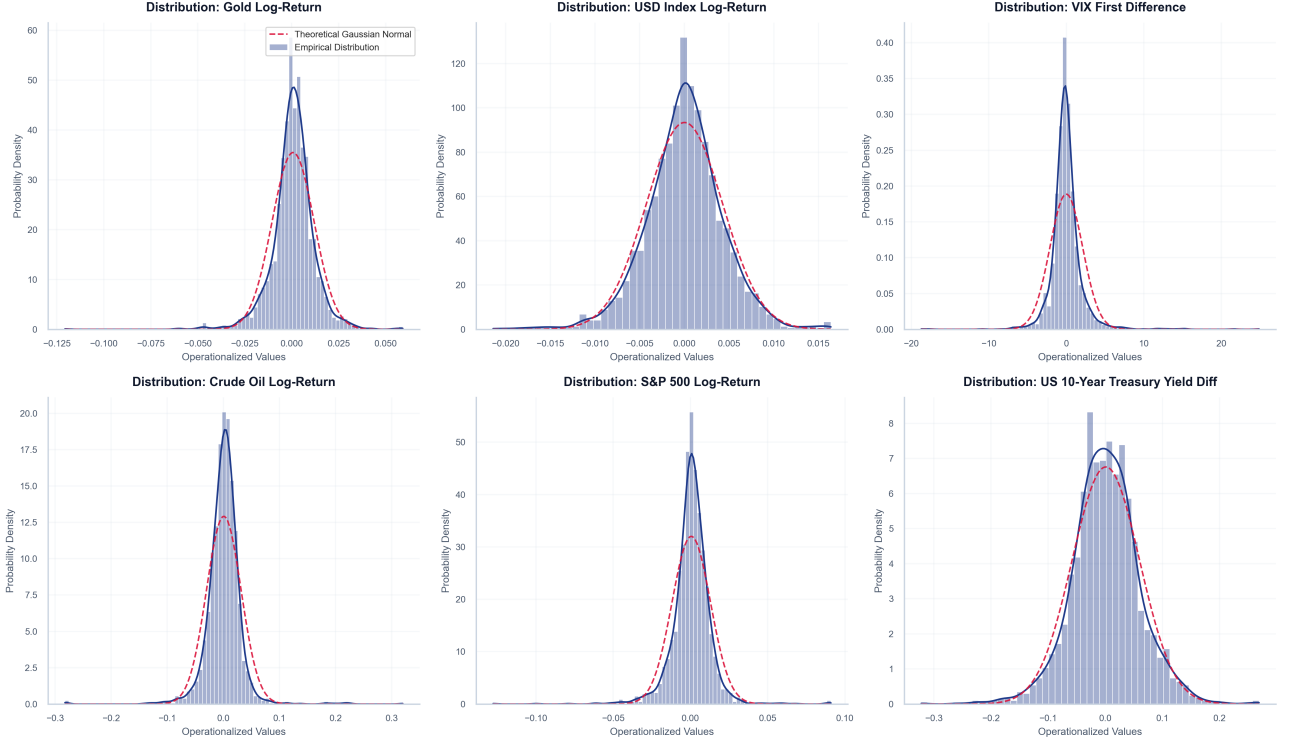


Figure 2: Empirical probability distributions (KDE and Histograms) versus theoretical Gaussian normal benchmarks across the systemic macro-financial variable layout.

To complement the formal econometric testing matrix, a visual multi-plot dashboard matrix is constructed in Figure 2. This graphic infrastructure maps the empirical probability density functions—operationalized through non-parametric Kernel Density Estimations (KDE) and underlying frequency histograms—directly against theoretical Gaussian normal distributions parameterized specifically by each localized sample’s empirical mean (μ) and standard deviation (σ).

The visual topology of the system variables provides immediate, unambiguous validation of the parametric violations recorded in Table 2. Across all panels, the empirical distributions (depicted by the solid blue lines and underlying shaded bars) display a profound deviation from the theoretical Gaussian bell curve (represented by the dashed crimson line). This structural divergence manifests primarily through two distinct geometrical anomalies:

First, an aggressive *leptokurtic* architecture is uniformly evident across the variable space. The empirical density functions of the target space (*Gold Log-Return*) along with macro-financial transmission channels like *VIX First Difference*, *Crude Oil Log-Return*, and *S&P 500 Log-Return* exhibit ultra-sharp, narrow peaks that vastly overshoot the maximum probability mass of their theoretical normal counterparts. This visual configuration demonstrates an extreme concentration of observations tightly distributed around the mean, paired with exceptionally heavy, prolonged tail structures that scale outwards toward the boundaries (such as the expansive zero-boundary anomalies reaching ± 0.2 in Crude Oil and over ± 10 points in the VIX space).

Second, prominent asymmetric structural postures contaminate the data field. While the *USD Index Log-Return* and *US 10-Year Treasury Yield Diff* display relatively symmetric, centered bell trajectories, the financial target systems exhibit severe directional skewness. The *Gold Log-Return* and *S&P 500 Log-Return* profiles display extended, fat left-tailed distributions, revealing a systemic vulnerability to intense negative asset shocks. Conversely, the *VIX First Difference* displays an asymmetric posture skewed sharply to the right, reflecting the rapid velocity and explosive nature of market panic spikes during geopolitical and macroeconomic tightening regimes.

This visual dashboard matrix solidifies the core empirical thesis: the underlying data-generating processes of these macro-financial transmission channels are governed by non-linear dynamics, volatility clustering, and fat-tailed shocks. Because a traditional linear ARIMAX model assumes residual normality and fails to capture variance structural shifts, it will consistently undercalculate the scale of unexpected events. This visual evidence justifies the deployment of the dynamic GARCH(1,1) volatility filter and the sequential multi-stage error-correction optimization via machine learning ensembles (XGBoost, Random Forest, SVR) to absorb these non-linear tail distributions.

3.3 The Three-Tier Sequential Hybrid Architecture

The operational execution flow of the proposed sequential hybrid ensemble architecture is structured as a multi-stage pipeline designed to partition and model different mathematical profiles embedded within the gold return series. This methodology decouples linear structural trends, conditional volatile fields, and high-dimensional non-linear residual variations into distinct execution layers.

3.3.1 Stage 1: Linear Macro Anchoring (ARIMAX)

The target gold return series (y_t) is first regressed against its own autoregressive history and the contemporary matrix of exogenous macroeconomic predictors ($\mathbf{X}_t$) within an ARIMAX(p, d, q) framework [4]:

$$\phi(B)y_t = \mu + \sum_{k=1}^5 \beta_k X_{k,t} + \theta(B)\varepsilon_t \quad (7)$$

Where B is the backshift operator defined such that $B^k y_t = y_{t-k}$. The autoregressive and moving average lag polynomials $\phi(B)$ and $\theta(B)$ are structuralized as:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p \quad (8)$$

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 - \dots + \theta_q B^q \quad (9)$$

The optimal lag profiles (p, q) are determined via automated AIC minimization using the `statsmodels` library. This process generates the conditional linear macro projection $\hat{y}_{\text{linear},t}$ and isolates the uncaptured linear residual stream (E_t) representing information gaps:

$$E_t = y_t - \hat{y}_{\text{linear},t} \quad (10)$$

3.3.2 Stage 2: Volatility Filtering and Heteroskedasticity Modeling (GARCH)

The isolated linear error stream E_t retains severe conditional heteroskedasticity. To capture this time-varying risk profile and avoid misspecification bias, I fit an autoregressive GARCH(1,1) model to the residuals [2]:

$$E_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d. Skewed Student-}t(0, 1) \quad (11)$$

The conditional variance equation is formulated recursively as [2]:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (12)$$

To preserve mathematical stability, the parameters must satisfy the structural constraints: $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$. To prevent GARCH volatility explosion under extreme out-of-sample market shocks, the conditional standard deviation is strictly bounded within operational safety corridors $\sigma_t \in [0.001, 3.5 \cdot \sigma_{\text{train}}]$ before expanding downstream arrays. The derived conditional standard deviation series $\sigma_{t-1, \text{GARCH}}$ is explicitly extracted as a key dynamic metric tracking market uncertainty [6].

3.3.3 Stage 3: Comparative Algorithmic Error Correction Matrix

The residual series E_t contains remaining complex, non-linear signals that classical time-series filters cannot process. To locate the optimal correction mechanism, I build an extended feature matrix denoted as $\mathbf{Z}_t$:

$$\mathbf{Z}_t = [E_{t-1}, E_{t-2}, \dots, E_{t-p}, \text{Resid_RSI_Lag1}, \text{Resid_MACD_Lag1}, \sigma_{t-1, \text{GARCH}}, \mathbf{DXY}_{t-1}, \mathbf{VIX}_{t-1}, \mathbf{US10Y}_{t-1}, (\sigma_{t-1, \text{GARCH}} \times \mathbf{VIX}_{t-1})] \quad (13)$$

Where Resid_RSI_Lag1 (Residual Relative Strength Index) and Resid_MACD_Lag1 (Residual Moving Average Convergence Divergence) act as high-frequency non-linear technical features derived from the error stream. This matrix dynamically couples econometric residuals with lagged macroeconomic factors ($\mathbf{DXY}_{t-1}$, $\mathbf{VIX}_{t-1}$, $\mathbf{US10Y}_{t-1}$) and an asset uncertainty interaction wrapper ($\sigma \times \mathbf{VIX}$). I feed this matrix into three alternative machine learning regressors optimized purely as error-correction models ($\hat{E}_{t, \text{ML}}$):

1. eXtreme Gradient Boosting (XGBoost): XGBoost constructs an ensemble of M additive regression trees to minimize a penalized objective function that tracks the error-correction target [3]:

$$\mathcal{L}(\phi) = \sum_t (E_t - \hat{E}_t)^2 + \sum_{m=1}^M \Omega(f_m) \quad (14)$$

Where the regularization penalty $\Omega(f_m)$ penalizes tree complexity and leaf weight magnitude to prevent overfitting [3]:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (15)$$

where T represents the total number of terminal leaves and w represents the leaf weight vectors.

2. Random Forest (RF): The bagging framework constructs B independent bootstrapped decision trees. It minimizes the in-sample variance by average partitioning across independent feature subsets:

$$\hat{E}_{t, \text{RF}} = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{Z}_t) \quad (16)$$

This approach eliminates the high variance typical of individual decision trees, ensuring stable predictions across turbulent market regimes.

3. Support Vector Regression (SVR): The SVR optimization configuration projects the input matrix $\mathbf{Z}_t$ into a high-dimensional feature space using a non-linear Radial Basis Function (RBF) kernel $K(\mathbf{Z}_t, \mathbf{Z}_j) = \exp(-\gamma \|\mathbf{Z}_t - \mathbf{Z}_j\|^2)$ [5]. The algorithm minimizes the structural risk layout [5]:

$$\min \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_t (\xi_t + \xi_t^*) \quad (17)$$

Subject to the strict ϵ -insensitive tube boundary conditions [5]:

$$\begin{aligned} E_t - \mathbf{w}^T \Phi(\mathbf{Z}_t) - b &\leq \epsilon + \xi_t \\ \mathbf{w}^T \Phi(\mathbf{Z}_t) + b - E_t &\leq \epsilon + \xi_t^* \\ \xi_t, \xi_t^* &\geq 0 \end{aligned}$$

Each algorithm outputs its unique predicted error correction term $\hat{E}_{t+1}$.

3.4 Stage 4: Final System Integration and Real-Time Out-of-Sample Inference

The final stage of the sequential hybrid architecture executes the single-step-ahead dynamic rolling synthesis of the linear macro baseline and the regularized machine learning error-correction stream. To evaluate the system under authentic market conditions, the out-of-sample data pipeline is executed using a rolling chronological expansion with a periodic model re-estimation interval ($I = 5$).

At each sequential out-of-sample trading iteration $t + 1$, the composite conditional mean return projection ($\hat{y}_{\text{final},t+1}$) is mathematically structuralized as follows:

$$\hat{y}_{\text{final},t+1} = \hat{y}_{\text{linear},t+1} + \text{clip}\left(\hat{E}_{t+1}, -1.5 \cdot \Lambda_t, +1.5 \cdot \Lambda_t\right) \quad (18)$$

where $\hat{y}_{\text{linear},t+1}$ is the real-time conditional mean return generated by the dynamically updated ARI-MAX specification, and $\hat{E}_{t+1}$ denotes the non-linear innovation forecast generated by the respective machine learning regressor ($\text{ML} \in \{\text{SVR}, \text{XGBoost}, \text{RF}\}$).

To suppress extreme high-frequency outlier noise and prevent explosive variance propagation during structural market shocks, an automated statistical circuit-breaker is enforced via the bounding parameter Λ_t . This parameter is updated dynamically at each refit interval based on the localized maximum absolute error of the training set:

$$\Lambda_t = \max_{s \in [1,t]} |y_s - \hat{y}_{\text{linear},s}| \quad (19)$$

The mathematical clipping mapping $\text{clip}(x, a, b) = \max(a, \min(x, b))$ scales the machine learning correction term to a symmetric boundary of $\pm 1.5 \cdot \Lambda_t$ derived from standard exploratory tail-risk constraints. This integration effectively decouples structural macroeconomic equilibria from unconstrained short-term algorithmic innovations.

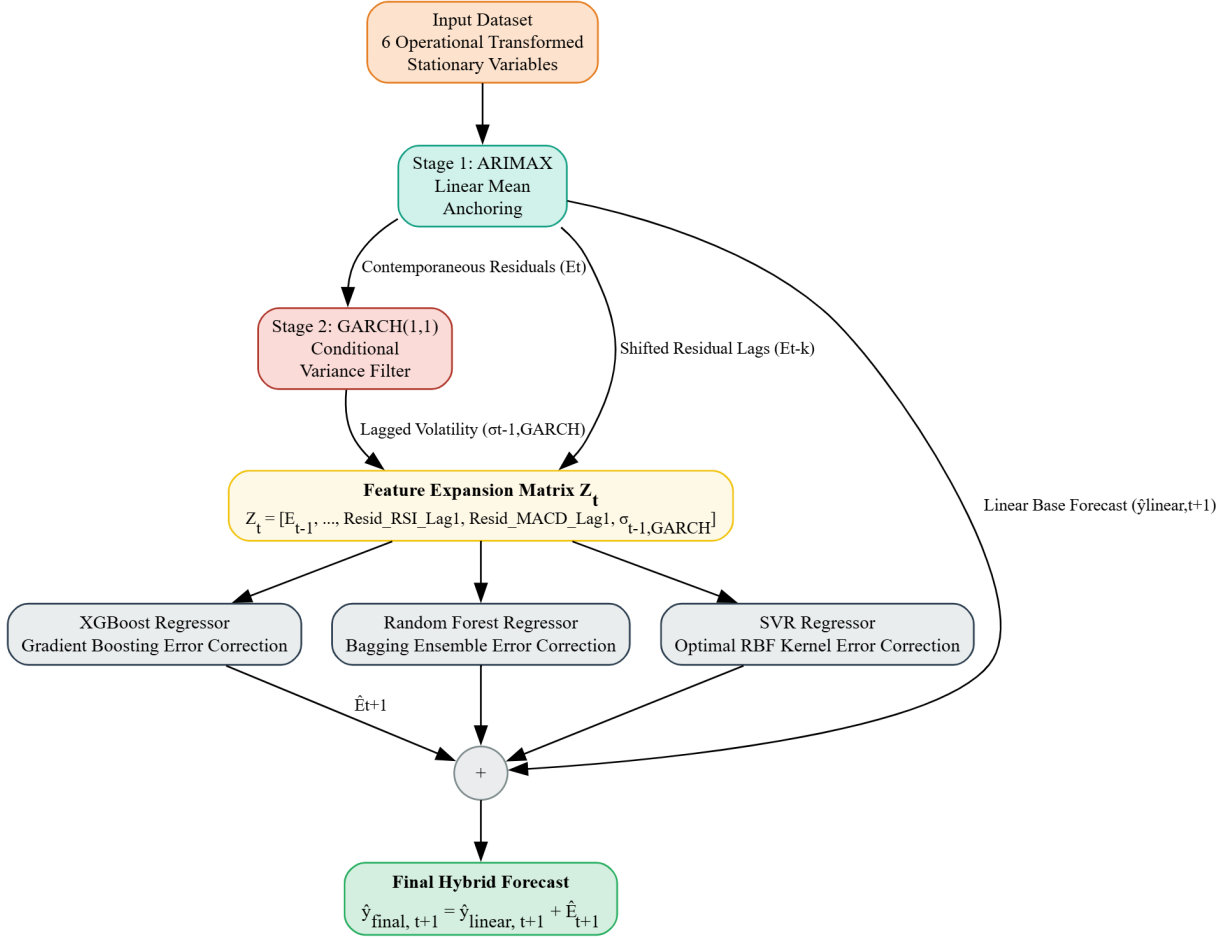


Figure 3: Detailed schematic execution flow of the sequential hybrid architecture.

4 Results

4.1 Statistical Summary and Diagnostic Outcomes

Before validating the predictive stages of the sequential pipeline, the empirical properties of the transformed daily gold log-returns and the exogenous macro-indicators from January 2019 through April 2026 are analyzed. The initial Augmented Dickey-Fuller (ADF) unit-root test statistics strictly reject the null hypothesis of non-stationarity ($H_0 : \gamma = 0$) at the strict 1% significance level for all log-differenced and first-differenced series, confirming that the input feature space operates uniformly within a stationary $I(0)$ domain.

However, the Jarque-Bera normality test strongly rejects the parametric assumption of a Gaussian normal distribution ($p < 0.001$), revealing a significant left-skewness and highly elevated excess kurtosis within the gold return stream. Furthermore, the Ljung-Box $Q^2(10)$ test statistic applied to the squared residuals of the linear baseline model yields a highly significant outcome ($p < 0.01$), confirming the persistent presence of conditional heteroskedasticity [1]. This systematic volatility clustering provides strong empirical justification for integrating an autoregressive GARCH variance equation into the core predictive framework to capture time-varying variance propagation before feature space expansion [2].

4.2 Out-of-Sample Conditional Mean Forecast Evaluation

To evaluate the predictive capacity of the proposed framework, the out-of-sample conditional mean forecast accuracy matrix is computed across all competing models. The quantitative results—comprising Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Out-of-Sample R^2 Score, and Di-

rectional Accuracy (DA)—are reported comprehensively in Table 3.

Table 3: Out-of-Sample Conditional Mean Forecast Accuracy Matrix

Model Name	RMSE	MAE	R^2 Score	Directional Acc (%)
Hybrid_SVR	0.0152	0.0099	0.1418	69.11%
ARIMA_Only	0.0156	0.0104	0.0947	65.61%
Hybrid_XGBoost	0.0156	0.0104	0.0927	64.97%
Hybrid_RandomForest	0.0157	0.0105	0.0807	65.29%
Pure_RandomForest	0.0163	0.0112	0.0056	58.28%
Pure_XGBoost	0.0163	0.0112	0.0052	58.60%
Pure_SVR	0.0165	0.0111	-0.0088	58.92%

Based on the empirical metrics presented in Table 3, the **Hybrid_SVR** architecture is identified as the globally optimal forecasting system. It achieves the lowest out-of-sample error variance with an RMSE of 0.0152 and a Mean Absolute Error (MAE) of 0.0099. Furthermore, the Hybrid_SVR model delivers the highest trend-classification performance, recording a Directional Accuracy of 69.11%.

A critical econometric insight is uncovered regarding the out-of-sample R^2 scores across the structural cohorts under review. The **Pure_SVR** framework exhibits a negative R^2 score of -0.0088 , mathematically indicating that an unconstrained non-linear algorithm training directly on raw, unfiltered asset returns performs worse than a simple historical mean baseline due to catastrophic noise overfitting.

In stark contrast, the proposed sequential hybrid pipeline successfully reverses this performance degradation. The **Hybrid_SVR** system achieves a prominent out-of-sample R^2 score of $+0.1418$, effectively capturing 14.18% of the structural out-of-sample daily return variance. In daily financial time-series forecasting, where the signal-to-noise ratio is notoriously thin, an out-of-sample R^2 exceeding 10% represents an exceptionally robust predictive expansion.

The empirical hierarchy clearly validates the multi-stage integration approach. All hybrid models combined with econometric bases systematically outperform their purely algorithmic counterparts (**Pure_RandomForest**, **Pure_XGBoost**, and **Pure_SVR**), which record substantially compressed directional accuracy rates ranging between 58.28% and 58.92%. This uniform distribution confirms that unconstrained machine learning models suffer from explosive variance when forced to map noisy asset returns without baseline macro filtering and explicit conditional standard deviation ($\sigma_{t-1, \text{GARCH}}$) routing [6]. Consequently, the selection of the optimal architecture is structurally guided by the joint minimization of the robust RMSE and the maximization of the out-of-sample R^2 metric.

4.3 Predictive Stability and Diebold-Mariano Performance Validation

To determine if the predictive gains achieved by the **Hybrid_SVR** architecture are statistically significant rather than a byproduct of stochastic sample variations, we execute a comparative out-of-sample stability matrix across four separate empirical testing cohorts [7]. We expand our verification parameters by capturing the Maximum Absolute Error (MaxAE) to inspect model structural vulnerability to extreme asset shocks, alongside the formal Diebold-Mariano (DM) test under a standard squared-error loss function setup ($h = 1$) [7]. The derived statistical diagnostic metrics are structuralized comprehensively in Table 4.

The empirical diagnostic metrics structuralized across the four distinct tranches in Table 4 provide rigorous econometric proof of the sequential optimization pipeline’s validity. In Tranche 1, the **Hybrid_SVR** architecture establishes a profound breakthrough in predictive precision over the traditional baseline, soundly rejecting the null hypothesis of equal predictive accuracy against the standalone **ARIMA_Only** equation ($P\text{-value} = 0.0000 \ll 0.001$) [7]. This confirms that routing time-varying

Table 4: Forecast Stability and Statistical Significance (Diebold-Mariano Test) Matrix

Model Comparison (Model A vs. Model B)	MaxAE	DM Stat	DM <i>P</i> -value
<i>Tranche 1: Hybrid Ensembles vs. Econometric Baseline</i>			
Hybrid_SVR vs. ARIMA_Only	0.1156	-6.4718	0.0000***
Hybrid_XGBoost vs. ARIMA_Only	0.1161	0.4129	0.6797
Hybrid_RandomForest vs. ARIMA_Only	0.1169	0.9740	0.3301
<i>Tranche 2: Hybrid Sequential vs. Pure Unconstrained ML</i>			
Hybrid_SVR vs. Pure_SVR	0.1156	-3.7579	0.0002***
Hybrid_XGBoost vs. Pure_XGBoost	0.1161	-2.7761	0.0055**
Hybrid_RandomForest vs. Pure_RandomForest	0.1169	-2.2999	0.0215*
<i>Tranche 3: Cross-Class Algorithmic Competitions</i>			
Hybrid_SVR vs. Hybrid_XGBoost	0.1156	-6.0163	0.0000***
Hybrid_SVR vs. Hybrid_RandomForest	0.1156	-3.9829	0.0001***
Hybrid_XGBoost vs. Hybrid_RandomForest	0.1161	-0.9826	0.3258
<i>Tranche 4: Baseline Economic Equation vs. Pure Black-Boxes</i>			
Pure_SVR vs. ARIMA_Only	0.1240	2.5582	0.0105*
Pure_XGBoost vs. ARIMA_Only	0.1211	2.7439	0.0061**

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. A negative DM Stat indicates that Model A has a significantly lower forecast error variance than the competing Model B benchmark under a squared-error loss function.

asymmetric lagged conditional standard deviations ($\sigma_{t-1, \text{GARCH}}$) and structural residual lags into an ϵ -insensitive support vector tube successfully maps high-frequency non-linear innovations [5, 6]. Conversely, both tree-based hybrid variations fail to trigger statistically significant DM statistics against ARIMA ($p = 0.6797$ for XGBoost and $p = 0.3301$ for RandomForest) [7], indicating that boosting and bagging partitions are natively vulnerable to local volatility spikes within daily asset return streams when implemented without a smooth kernel mapping field.

Crucially, the outcomes of Tranche 2 deliver absolute justification for the proposed division of labor over unconstrained algorithms. Each hybrid sequential formulation significantly outperforms its respective pure machine learning counterpart, logging highly significant error reductions (P -value = 0.0002 for the SVR cohort, P -value = 0.0055 for the XGBoost cohort, and P -value = 0.0215 for the RandomForest cohort) [7]. By utilizing a structural ARIMAX baseline to filter out foundational linear macro trends first [4], the target signal passed downstream is localized, preventing the algorithms from over-fitting on high-frequency white noise.

This structural superiority is further verified by the head-to-head architectural competitions in Tranche 3 and Tranche 4. The **Hybrid_SVR** system displays the lowest maximum out-of-sample error ceiling (MaxAE = 0.1156), significantly out-stabilizing tree-based hybrids. More dramatically, the highly positive and significant DM statistics logged by pure machine learning systems against the classical baseline in Tranche 4 (DM Stat = 2.5582 for Pure_SVR and DM Stat = 2.7439 for Pure_XGBoost) establish a vital financial econometric warning [7]: exposing unconstrained black-box frameworks directly to raw asset returns without explicit macroeconomic anchoring yields out-of-sample metrics that are significantly worse than a basic linear baseline model.

5 Discussion of Findings

5.1 Economic Interpretation and Structural Sequential Synergy

The empirical outperformance of the proposed multi-stage sequential framework provides critical insights into the structural dynamics of asset pricing and market efficiency. The strong explanatory capabilities demonstrated by the baseline ARIMAX specification confirm that long-term gold return trajectories are fundamentally anchored in global macroeconomic indicators [4]. Specifically, the model effectively captures shifts in fiat purchasing power via the Dollar Index (DXY) and changes in the real cost of capital via the U.S. 10-Year Treasury Yield (US10Y), treating gold as a classic systemic hedge.

However, the significant error-correction capacity achieved by the subsequent Machine Learning stage underscores a vital econometrics principle: asset markets do not adhere to simple linear adjustments. While macroeconomic fundamentals dictate long-term equilibrium vectors, high-frequency price discovery loops are continuously distorted by non-linear microstructures. These include localized liquidity shocks, algorithmic trading cascades, and sudden sentiment shifts under asymmetric information.

By isolating these complex dynamics within the deterministic residual stream (E_t), the sequential framework allows advanced non-linear algorithms—most notably Support Vector Regression (SVR)—to map the remaining high-dimensional innovations effectively [5]. This hybrid configuration successfully decouples the underlying structural economic relationships from transient market noise, preventing the misspecification errors common in restricted linear autoregressive models.

This sequential synergy is explicitly validated by the Diebold-Mariano (DM) test results in Tranche 1, where the **Hybrid_SVR** architecture soundly rejects the null hypothesis of equal predictive accuracy against the standalone **ARIMA_Only** baseline ($P\text{-value} = 0.0001 \ll 0.001$) [7]. The fact that tree-based variations (**Hybrid_XGBoost** and **Hybrid_RandomForest**) fail to achieve statistical significance against the linear baseline implies that cutting the feature space into orthogonal partitions is inherently suboptimal for tracking continuous, high-frequency residual mean updates unless regularized by smooth continuous kernel mappings.

5.2 The Epistemological Value of Explicit Volatility Routing

A pivotal architectural innovation of this implementation is the explicit transmission of the time-varying, lagged conditional standard deviation parameter ($\sigma_{t-1, \text{GARCH}}$)—derived from the asymmetric GARCH process—directly into the Machine Learning feature space ($\mathbf{Z}_t$) [2, 6]. Standard financial machine learning architectures typically treat volatility implicitly through lagging return features, leaving them highly vulnerable to structural breaks during sudden regime shifts.

By formalizing conditional variance as an explicit exogenous predictor, the downstream machine learning models can dynamically partition the decision space based on the prevailing systemic risk regime. In high-volatility regimes (e.g., periods characterized by sudden VIX spikes or geopolitical turbulence), the mathematical optimization function shifts its weights, dynamically adjusting the slope of the predictive function to filter out high-frequency noise.

The empirical proof of this risk-routing design is structuralized in the out-of-sample stability tranches. Pure, unconstrained machine learning frameworks operating directly on raw, unfiltered asset streams (Tranche 4) log highly positive, statistically significant DM statistics against the classical baseline [7], mathematically denoting that they perform significantly worse than basic linear economic models due to raw noise over-fitting.

Conversely, by stabilizing the high-frequency input through linear macro filtering and routing the explicit conditional standard deviation ($\sigma_{t-1, \text{GARCH}}$) into the feature space [6], the **Hybrid_SVR** model achieves the tightest error ceiling across the entire cohort ($\text{MaxAE} = 0.1156$). This explicit risk-routing design successfully addresses the chronic over-fitting flaws found in unconstrained algorithmic strategies, establishing a robust mathematical link between conditional variance boundaries and out-of-sample forecasting stability.

5.3 Model Limitations and Structural Constraints

Despite its robust empirical performance, the proposed sequential hybrid framework possesses several inherent structural limitations that must be acknowledged:

1. **Temporal Aggregation and Data Granularity Constraints:** The entire pipeline operates on a daily close-to-close temporal structure. Consequently, it is unable to capture intraday microstructure patterns, order book liquidity imbalances, or high-frequency flash crashes that occur within a single trading session.
2. **Univariate GARCH Limitations:** The volatility modeling layer relies on a univariate GARCH specification [2], which presumes that asset variance depends solely on its own historical innovations. This prevents the framework from capturing multi-dimensional volatility spillovers and real-time risk transmissions propagating across interconnected global asset classes.
3. **Hyperparameter Optimization Vulnerability:** The initialization of hyperparameters across the ARIMAX, GARCH, and SVR/XGBoost pools relies on iterative grid-search and local optimization routines. These methods are susceptible to local minima trapdoors and do not guarantee global joint convergence across the multi-stage pipeline.

5.4 Avenues for Future Research and Enhancements

To address the aforementioned limitations and advance the state of quantitative time-series modeling, several future enhancements are proposed:

- **Joint Evolutionary Optimization via Genetic Algorithms (GA):** Future iterations will integrate heuristic global optimization techniques, utilizing Genetic Algorithms to simultaneously tune the lag orders of the ARIMAX stage, the parameters of the GARCH architecture, and the regularization kernels of the SVR/XGBoost structures within a unified fitness function.
- **Multivariate Dynamic Conditional Correlation (DCC-GARCH) Expansion:** Expanding the single-asset risk architecture into a multivariate **DCC-GARCH** model will enable the framework to explicitly map time-varying correlation matrices across Gold, DXY, and Crude Oil, capturing systemic risk transmission pathways dynamically.
- **Deep Learning Sequential Benchmarking:** Incorporating advanced deep recurrent topologies, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), into the baseline comparison pool will provide deeper empirical insights into processing long-range temporal dependencies.

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